

SARAPUL, Russian Federation

WMO: 284180

Lat: 56.467N

Lon: 53.733E

Elev: 133

StdP: 99.74

Time Zone: 4.00 (E04)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.4	-24.6	-30.2	0.2	-27.2	-27.3	0.3	-24.5	7.6	-8.6	6.5	-9.1	1.1	0	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	31.1	21.1	29.1	20.0	27.3	19.2	22.0	29.4	21.0	27.6	19.9	25.9	2.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.3	25.2	18.4	13.5	24.5	17.5	12.7	23.6	64.7	29.8	61.1	27.5	57.5	25.9	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.8	5.2	-31.4	32.8	3.5	2.0	-33.9	34.3	-36.0	35.5	-38.0	36.6	-40.5	38.1		
(5)			-31.5	23.3	3.5	1.0	-34.0	24.0	-36.1	24.6	-38.0	25.2	-40.6	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-13.1	-10.9	-4.5	4.8	13.6	17.6	19.5	18.0	11.8	3.9	-2.8	-9.1		
	DBStd	12.59	6.72	6.40	4.84	4.85	5.07	4.42	3.78	4.79	4.03	4.56	5.33	6.82		
	HDD10.0	3144	717	584	451	169	22	2	0	1	26	197	385	591		
	HDD18.3	5362	975	817	709	405	161	66	32	67	198	448	635	849		
	CDD10.0	1008	0	0	0	14	136	229	293	248	80	7	0	0		
	CDD18.3	183	0	0	0	0	16	43	67	55	3	0	0	0		
	CDH23.3	1703	0	0	0	2	166	378	615	522	21	0	0	0		
	CDH26.7	529	0	0	0	0	34	98	197	198	2	0	0	0		
(14) Wind	WSAvg	2.3	2.3	2.1	2.4	2.4	2.4	2.3	2.1	2.1	2.1	2.4	2.3	2.2		
(15) Precipitation	PrecAvg	549	38	28	26	31	44	60	59	64	54	54	46	42		
	PrecMax	731	77	86	89	88	121	123	149	184	133	105	107	116		
	PrecMin	322	7	0	0	6	2	12	0	8	1	4	14	2		
	PrecStd	91	16	16	19	19	27	32	31	36	33	24	21	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.7	6.7	23.1	29.8	31.5	33.8	34.4	26.1	17.4	8.8	2.2		
		MCWB	-0.2	0.5	3.0	14.8	18.7	21.3	22.8	22.7	16.7	11.4	7.8	1.6		
	2%	DB	-0.2	0.7	4.2	19.2	27.4	29.4	31.1	31.3	22.6	14.1	7.3	1.2		
		MCWB	-0.9	-0.2	1.5	12.4	17.7	20.8	21.6	20.7	15.5	10.0	6.3	0.5		
	5%	DB	-1.7	-0.5	2.8	15.4	25.0	27.5	28.8	28.7	20.3	12.1	5.7	0.4		
		MCWB	-2.6	-1.3	0.8	9.4	16.5	19.6	20.0	20.1	14.3	9.1	4.9	-0.2		
	10%	DB	-3.9	-2.6	1.7	12.7	22.7	25.4	26.8	26.0	18.4	10.2	3.2	-1.0		
		MCWB	-4.6	-3.5	0.3	8.2	15.1	18.4	19.3	18.9	13.5	8.2	2.5	-1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	0.9	4.0	15.3	20.0	22.6	23.4	23.0	17.7	12.7	8.1	1.7		
		MCDB	0.7	1.4	5.9	22.4	27.0	29.4	31.8	32.7	22.5	15.7	8.7	1.9		
	2%	WB	-0.9	0.0	2.1	12.6	18.5	21.5	22.1	21.6	16.4	10.7	6.7	0.7		
		MCDB	-0.2	0.6	3.6	17.8	26.0	28.2	29.6	29.3	21.0	13.0	7.4	1.1		
	5%	WB	-2.5	-1.5	1.3	10.2	17.1	20.4	21.0	20.5	15.2	9.5	4.8	-0.2		
		MCDB	-1.7	-0.6	2.3	14.9	24.2	26.4	27.0	27.4	19.0	11.6	5.6	0.4		
	10%	WB	-4.6	-3.4	0.4	8.2	15.5	19.0	19.9	19.3	14.1	8.3	2.5	-1.6		
		MCDB	-3.9	-2.6	1.6	12.4	21.7	24.2	25.5	25.3	17.4	10.3	3.1	-1.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.2	7.1	8.0	11.0	9.8	9.9	9.4	8.0	5.3	3.8	5.0		
		MCDBR	4.9	4.0	6.4	12.8	13.6	12.1	13.0	12.7	11.7	8.5	4.0	3.7		
		MCWBR	4.8	3.7	4.5	7.4	7.0	5.7	5.9	5.4	6.2	5.4	3.9	3.7		
		MCDBR	4.7	3.8	5.7	11.9	12.7	10.9	11.9	11.4	10.3	7.1	4.1	3.7		
	5% WB	MCWBR	4.7	3.6	4.5	7.1	7.0	5.3	6.0	5.4	5.8	5.1	4.1	3.7		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.318	0.373	0.405	0.392	0.388	0.409	0.429	0.376	0.353	0.313	0.290		
	taud		2.022	2.025	1.945	2.094	2.292	2.356	2.305	2.223	2.344	2.271	2.139	1.999		
	Ebn at Noon		550	705	755	802	846	854	825	774	769	673	533	440		
	Edh at Noon		64	97	136	137	120	114	118	120	92	77	58	51		
(44) All-Sky Solar Radiation	RadAvg		0.67	1.62	3.06	4.35	5.38	5.75	5.71	4.35	2.70	1.20	0.52	0.37		
	RadStd		0.09	0.21	0.29	0.45	0.52	0.49	0.55	0.40	0.31	0.23	0.10	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		+0.57	N/A	N/A	N/A	N/A	N/A	N/A	-201	N/A	N/A

Nomenclature: See separate page